



COLM 2026

**University
at Buffalo**

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AUROC on LLaDA-8B (LLM-as-Judge)

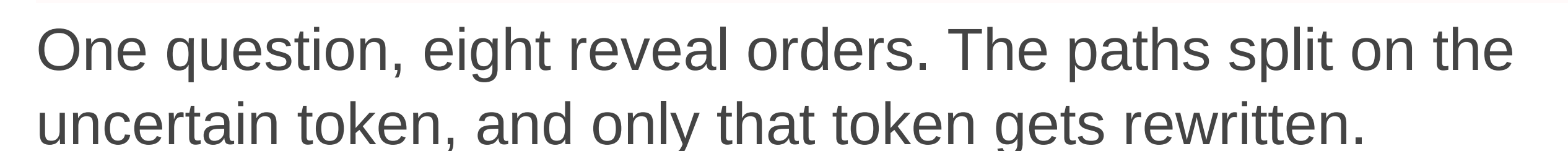
of hallucinations found in the top 20% of tokens

F1 after correction (LLaDA-8B)

runtime, no training

Motivation

- Run several chains in different reveal orders. Where they disagree, the model is guessing.



The OSCAR Pipeline



- 1 Decode 8 chains in parallel, each revealing tokens in a random order. They share weights and run in one batched pass per step.
- 2 At each position, measure how much the chains disagree (cross-chain entropy). Flag the top 20% and group neighbors into spans.
- 3 Keep the chain closest to the consensus, remask only the flagged spans, and denoise them for 8 more steps. A passage retrieved with the span text can guide the fill.

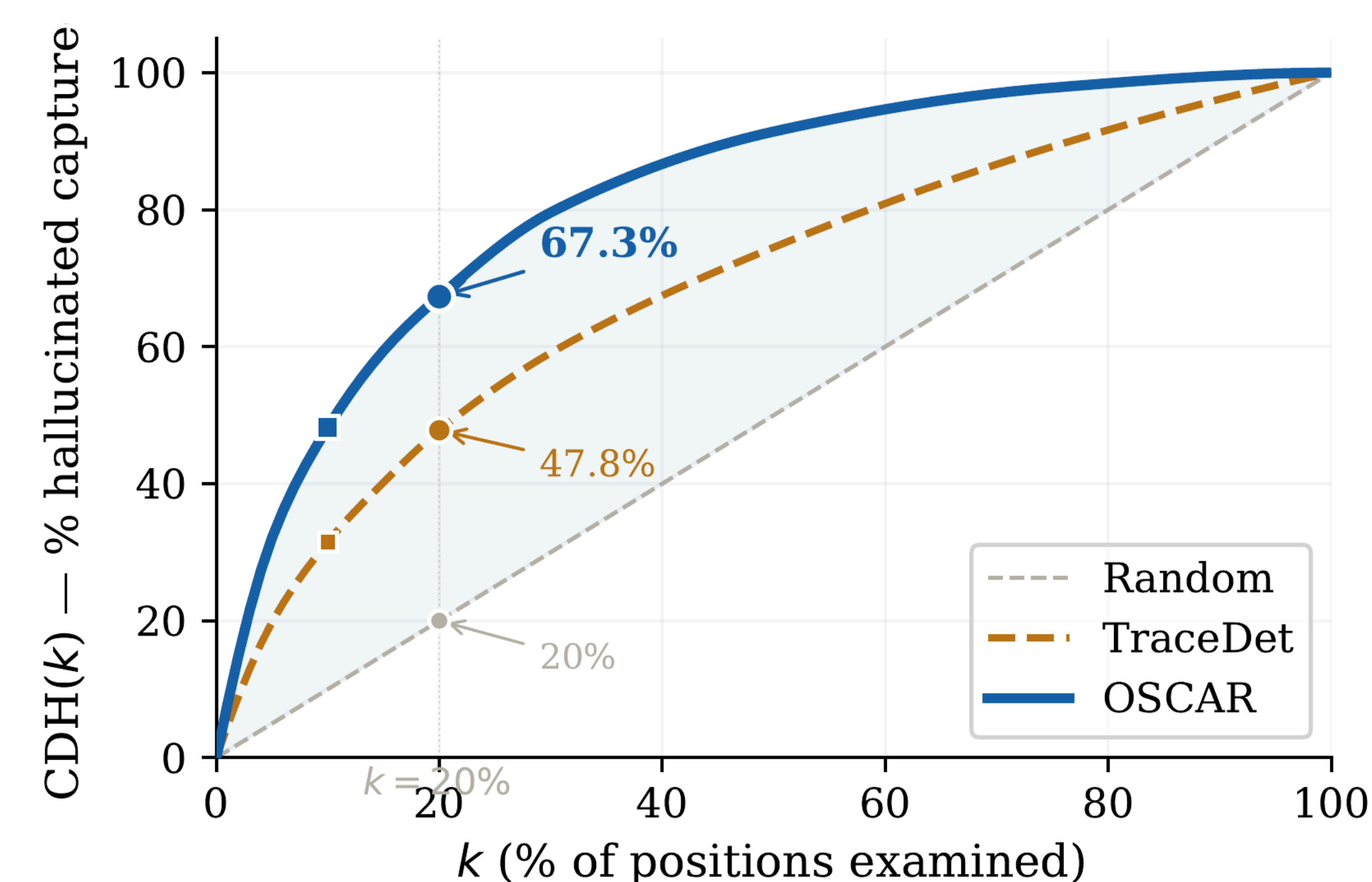
No training and no labels. Correction touches about 18.7% of tokens, and a full run takes about 1.3× as long.

Detection

With an LLM judge, the untrained OSCAR signal beats the best trained detector on both models.

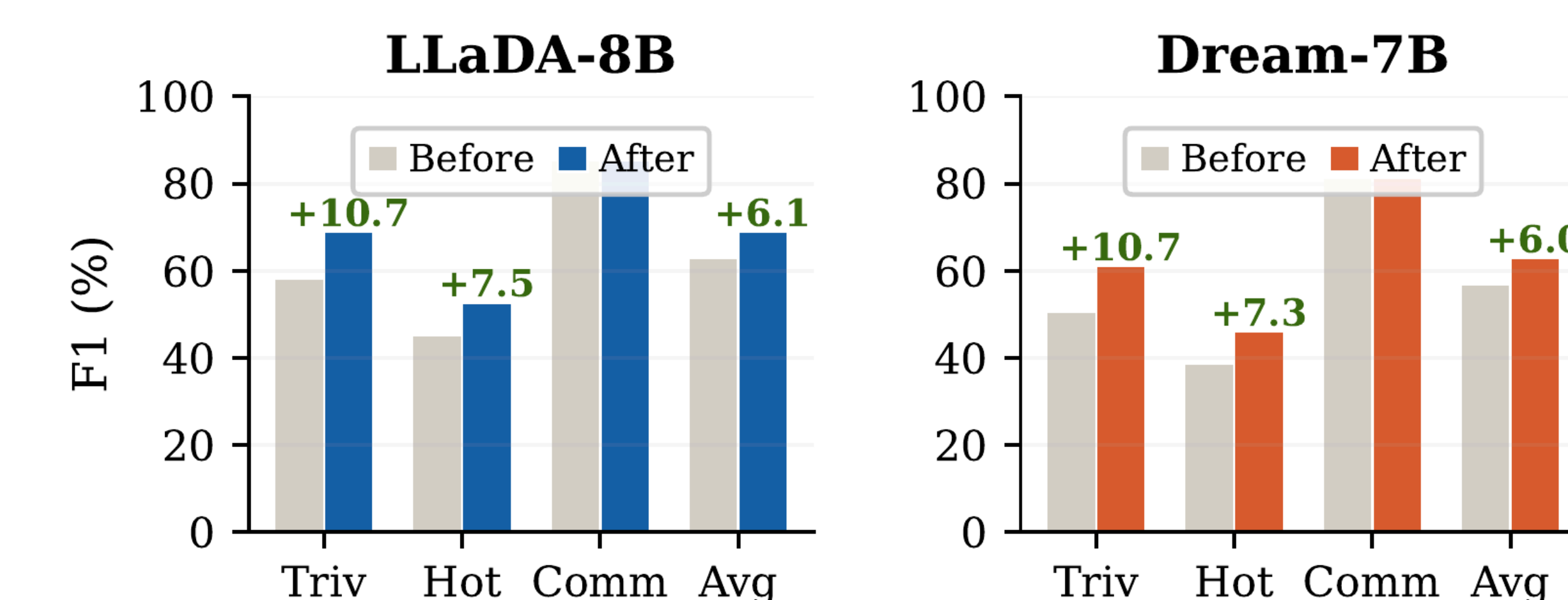
Averaged over TriviaQA, HotpotQA and CommonsenseQA. * needs a classifier trained on hallucination labels. With exact-match scoring, DynHD stays ahead.

- On TriviaQA, the 95% confidence interval (86.8–92.3) sits entirely above DynHD's 84.2.
- OSCAR beats the best SelfCheckGPT variant for DLMs by 18.7 AUROC points.



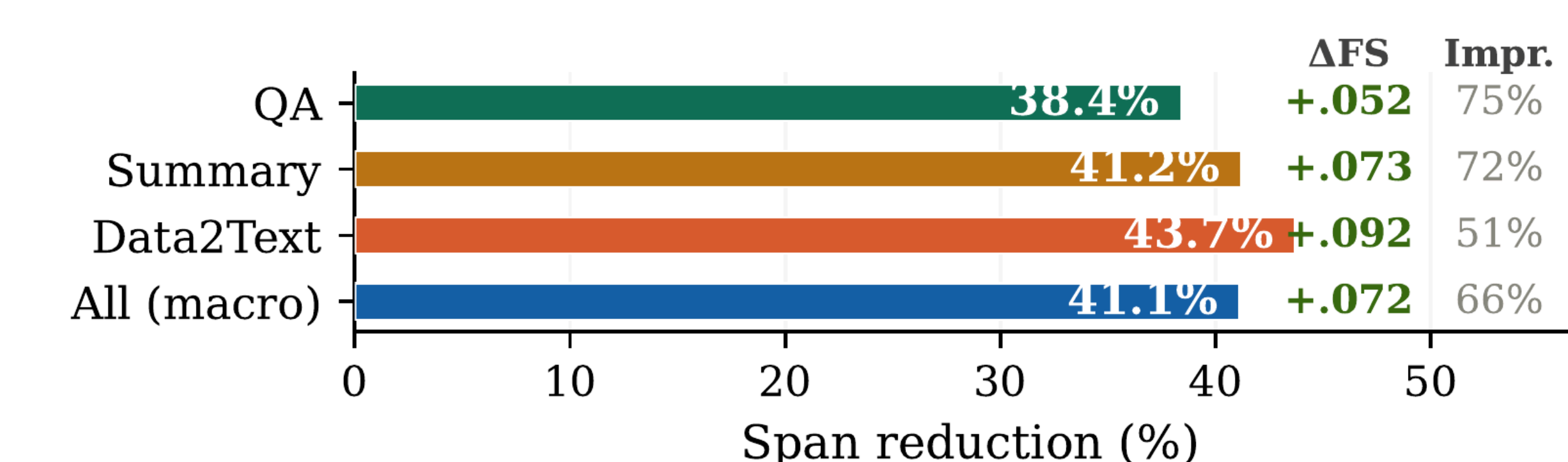
Where the errors hide: the 20% most uncertain tokens hold 67.3% of hallucinated positions (TraceDet: 47.8%).

Correction



Answers improve after correction, most on TriviaQA (+10.7 F1). CommonsenseQA stays flat because the chains already agree.

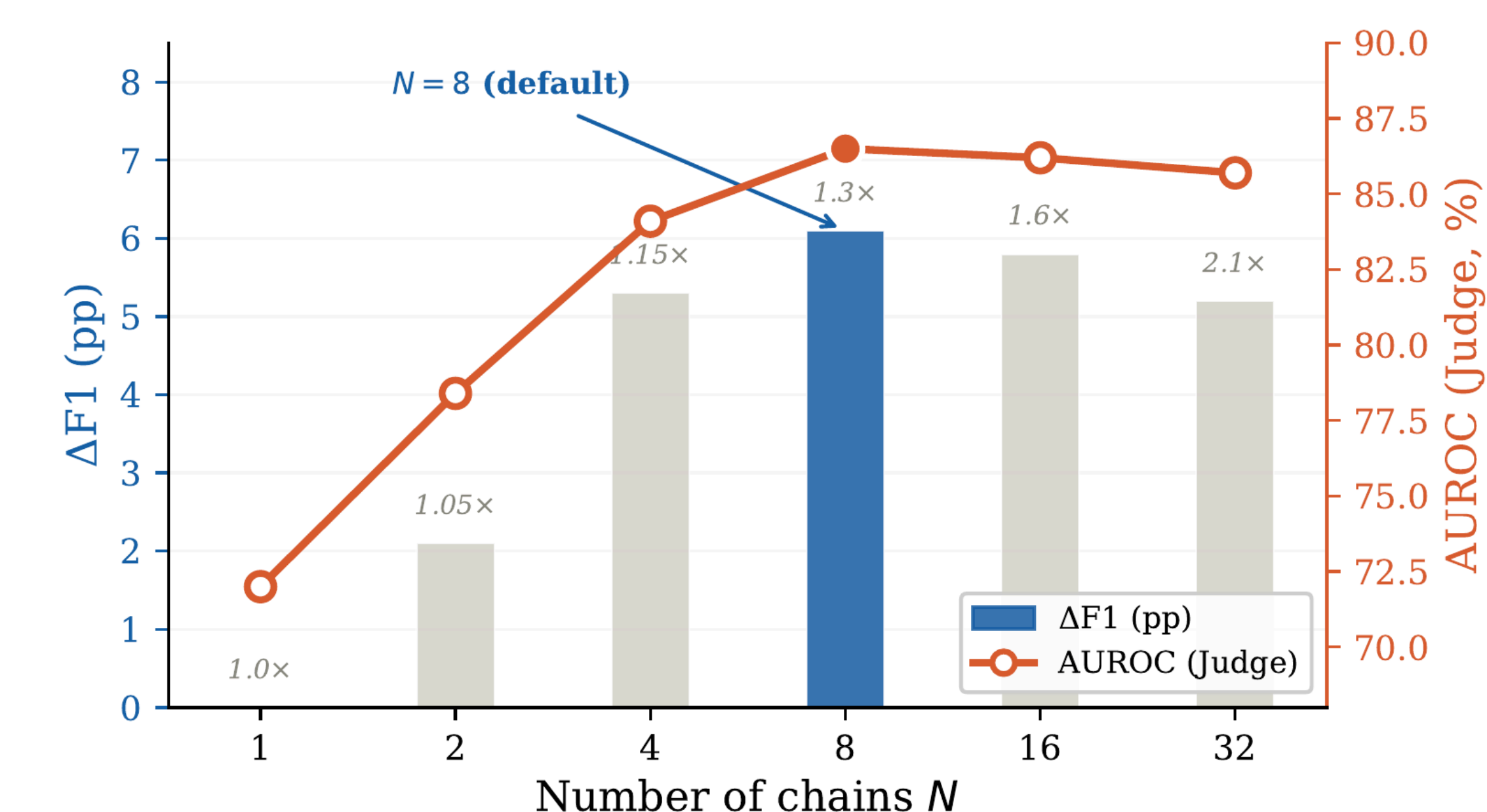
- Fixing only the flagged spans adds +6.1 F1. Random spans add nothing (-0.1), and untargeted fixes add +2.5.
- When OSCAR changes an answer, it usually helps: 91.9% of corrections improve TriviaQA and 97.1% improve HotpotQA.
- Retrieving a passage for each uncertain span adds another +3.4 F1.



It works on longer outputs too: on RAGTruth, hallucinated spans shrink 41.1% and FactScore rises in every subset.

Analysis

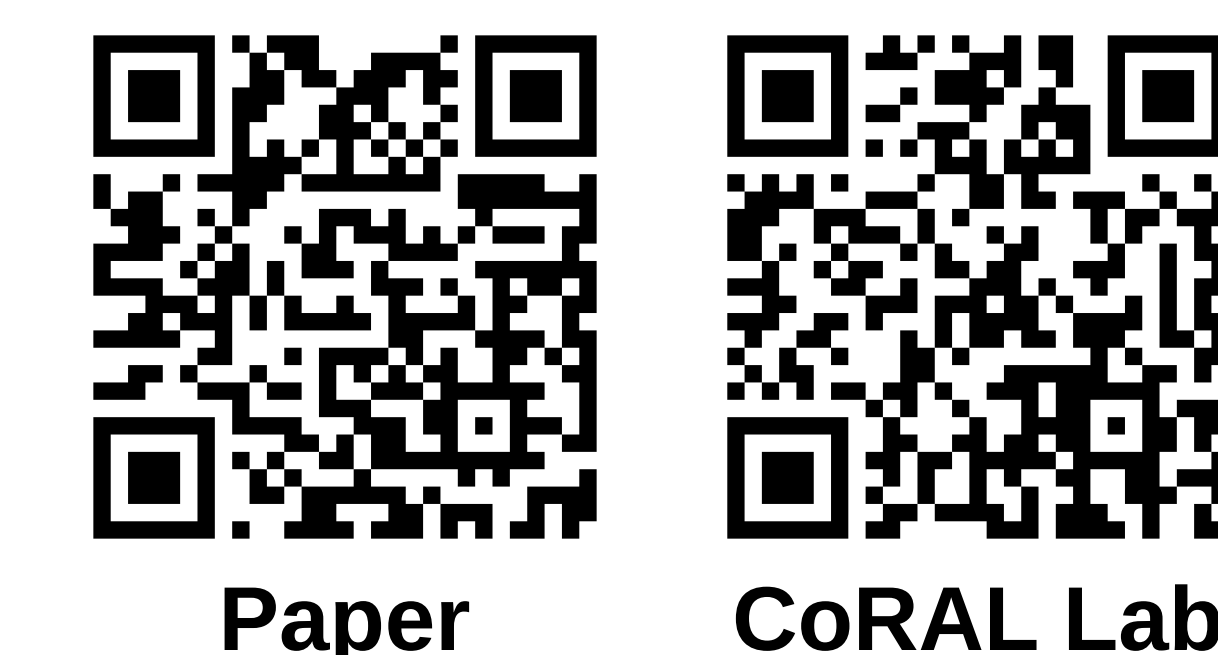
- DLMs carry their own hallucination signal. Autoregressive models pass only the top token forward, which hides it.
- Eight chains is the sweet spot; gains level off beyond it.
- One learned step followed by seven random ones gives 69.0 F1, against 61.8 when every step is learned.



More chains help up to 8, then the gains flatten out.

Limits

- 9–20% of hallucinations are confident and wrong. Those need retrieval, not remasking.
- We tested two models, LLaDA-8B and Dream-7B



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